

LABORATORY 2

Introduction to Assembly Programming

Objectives:

- To learn 8086 16-bit Intel Microprocessor, its register and assembly level programming.
- To learn the EMU8086 Emulator.
- To learn assembly programming by practicing simple programs including average calculation of 3/4/5/more numbers, calculate area of a rectangle and a triangle, temperature conversion from °C to °F, conversion from °F to °C, conversion from °C to °K, conversion from °K to °C and counting tiles problems.

Experimental tools:

MDA-Win8086, Computer, Microprocessor emulator Software with Integrated Assembler.

The following rules are needed to perform the lab work.

Instructions:

RES System Reset

AD Set Memory address

DA Update segment && offset.

STP Execute user's program, a single step.

GO Go to user's program or execute monitor functions.

MON Immediately break user's program and Non makeable interrupt.

REG Register display

+ Segment & offset +1 increment. Register display increment.

- Segment & offset -1 increment. Register display increment.

From the emulator you will get the **HEX** code for your assembly language program. For execute code and get the result first of all press **RES** button then press **DA** button, now type your **HEX** code here go to next address pressing **+** button. After completing the code typing you have to press **STP** button for executing the code. Now for watch result press **REG** button then you can see the result in the display of MDA-Win8086.

Instruction set:

MOV: This instruction allows copying the value of one register into another register.

ADD: This instruction adds two numbers.

SUB: This instruction subtracts a number from another number.

MUL: This instruction multiplies two numbers.

DIV: The instruction divides a number by another number.

AX= It is called accumulator register.

BX= It is called base register.

CX= It is called count register.

DX=It is called data register

Problem 1: Temperature conversion from °C to °K let, temperature = 39°C $1^{\circ}\text{K}=1^{\circ}\text{C} + 273$	
MOV AX, 39 MOV BX, 273 ADD AX, BX INT 3	
<u>Output:</u>	AX=0138 BX=0111 CX=0000 DX=0000
<u>Result Verification:</u>	K= 39+273= 312
<u>Discussion:</u> We know, $^{\circ}\text{K}=^{\circ}\text{C} + 273$ At first, 27 loaded in AX register and the address is 0404 and 111 replaced in BX register and its address is 0407. Now, AX, BX are added in address 041A. After pressing STP and REG, it shows the result. INT 3: INT 3 is a special one-byte instruction having op-code is CCH. that is inserted by debuggers at the instruction where the user has set a breakpoint to occur. When it's hit, the interrupt handler breaks into the debugger and then replaces the original instruction so that execution can proceed when the user is ready.	

Merge Problem 1 and Problem 2 and show the students about the task done by the INT3	
Problem 2: Temperature conversion from °K to °C let, temperature = 270°K	
MOV AX, 270 MOV BX, 273 SUB AX, BX INT 3	
<u>Output:</u>	AX=FFFD BX=0111 CX=0000 DX=0000
<u>Result Verification:</u>	C = 270-273 = -3 = FFFDH
<u>Discussion:</u> We know, °C = °K – 273 At first, 10E replaced in Ax register and the address is 0404 and 111 replaced in Bx register and its address is 0407. Now, Ax, Bx are subtract in address 040A. After pressing STP and REG, it shows the result.	

Problem 3: Average of 3 numbers: (2+3+5)/3	
MOV AX, 2 MOV BX, 3 ADD AX, BX MOV BX, 5 ADD AX, BX MOV BX, 3 DIV BL INT 3	
<u>Output:</u>	AX=0103 BX=0003 CX=0000 DX=0000
<u>Result Verification:</u>	Avg = (2+3+5)/3 = 3 AL = 3, AH = 1
<u>Discussion:</u> At first, load 2 in AX register and the address is 0404 and load 3 in BX register and its address is 0407. Now, AX, BX are added in 040A then load 5 in BX in 040C, and AX, BX are added	

again in 040F. Now, load 3 in BX and the address is 0411. Then BL is divided in 0413 address. After pressing STP and REG, it shows the result.

Problem 4: Average of 5 numbers: $((2+3+4+1+5)/5)$

```
MOV AX, 2
MOV BX, 3
ADD AX, BX
MOV BX, 4
ADD AX, BX
MOV BX, 1
ADD AX, BX
MOV BX, 5
ADD AX, BX
MOV BX, 5
DIV BL
INT 3
```

Output:

AX=0003 BX=0005
CX=0000 DX=0000

Result Verification:

Avg= $(2+3+4+1+5)/5 = 15/5$
AL = 3, AH = 0

Discussion:

At first, 2 replaced in Ax register and the address is 0404 and 3 replaced in Bx register and its address is 0407. Now, Ax, Bx are added in 040A then 4 replaced in Bx in 040C, and Ax, Bx are added again in 040F. Now, 1 replaced in Bx and the address is 0411. Then Bx is added in 0414 address and again 5 replaced in Bx and the address is 0416 then Bx is added in 0419. Now, 5 replaced in Bx and the address is 041B. Then Ax is divided by BL in 041D address. After pressing STP and REG, it shows the result.

Problem 5: Floor size 20*20, Tiles size 2*2. How many tiles are needed to cover up the floor?

```
MOV AX, 20
MOV BX, 20
MUL BL
MOV CX, AX
MOV AX, 2
```

MOV BX, 2 MUL BL MOV BX, AX MOV AX, CX DIV BL INT 3		
<u>Output:</u>	AX=0064 CX=0190	BX=0004 DX=0000
<u>Result Verification:</u>	Tiles = $(20*20)/(2*2) = 400/4 = 100 = 64H$	

Problem 6: Factorial Operation: 5! – 3!		
<pre>MOV AX, 1 MOV CL, 5 L1: MUL CL LOOP L1 MOV DX, AX MOV AX, 1 MOV CL, 3 L2: MUL CL LOOP L2 MOV BX, AX MOV AX, DX SUB AX, BX INT 3</pre>		
<u>Output:</u>	AX=0072 CX=0000	BX=0006 DX=0078
<u>Result Verification:</u>	5! – 3! = 114 = 72H; AH = 00, AL = 72	
<u>Discussion:</u> At first, we load 1 in AX register and load 5 in CL register then do multiply by giving loop with CL address and move AX value in DX register. Now, again entered value 1 in AX register and 3 replaced in CL register then do multiply by giving loop with CL address and		

move AX value in BX register. Then move the DX value in AX register and do subtraction of AX and BX. After pressing STP and REG, it produces the result.

Problem 7: $(5! / 3!) + 4!$

```
MOV AX, 1
MOV CL, 5
L1:MUL CL
LOOP L1
MOV DX, AX
MOV AX, 1
MOV CL, 3
L2:MUL CL
LOOP L2
MOV BX, AX
MOV AX, DX
DIV BL
MOV DX, AX
MOV AX, 1
MOV CL, 4
L3: MUL CL
LOOP L3
ADD AX,DX
INT 3
```

Output:

AX=002C
CX=0000

BX=0006
DX=0014

Result Verification:

$(5! / 3!) + 4! = (120/6) + 24 = 20 + 24 = 44 = 2C H$,
AH=00, AL=2C

Discussion:

At first, load 1 in AX register and load 5 in CL register then do multiply by giving loop with CL register and move AX value in DX register. Again, load value 1 in AX register and 3 in CL register then do multiply by giving loop with CL register and move AX value in BX register. Then move the DX value in AX register and do division by BL. Now, move AX value in DX and again entered value 1 in AX register and 4 replaced in CL register then do multiply by giving loop with CL address. At last, we do addition of DX and AX. After pressing STP and REG, we get the result.

Problem 8: (2! * 3! * 4!) + 4!

```
MOV AX, 1
MOV CL, 2
L1:MUL CL
LOOP L1
MOV DX, AX
MOV AX, 1
MOV CL, 3
L2:MUL CL
LOOP L2
MOV BX, AX
MOV AX, DX
MUL BL
MOV DX, AX
MOV AX, 1
MOV CL, 4
L3:MUL CL
LOOP L3
ADD AX, DX
INT 3
```

Output:

AX=0138
CX=0000

BX=0018
DX=0120

Result Verification:

$(2! * 3!) + 4! = (2 * 6) + 24 = 12 + 24 = 36 = 24$
HDH=00, DL=24

Discussion:

At first, load 1 in AX register and load 2 in CL register then do multiply by giving loop with CL address and move AX value in DX register. Again, enter value 1 in AX register and 3 in CL register then do multiply by giving loop with CL address and move AX value in BX register. Then move the DX value in AX register and do division by BL. Now, move AX value in DX and again load value 1 in AX register and 4 in CL register then do multiply by giving loop with CL address. Now, we do addition of DX and AX in address. At last, we do addition of DX and AX. After pressing STP and REG, we get the result.

Problem 9: (4!/2!)/3!

```

MOV AX, 1
MOV CL, 4
L1:
    MUL CL
LOOP L1
MOV DX, AX
MOV AX, 1
MOV CL, 2
L2:
    MUL CL
LOOP L2
MOV BX, AX
MOV AX, DX
DIV BL
MOV DX, AX
MOV AX, 1
MOV CL, 3
L3:
    MUL CL
LOOP L3
MOV BX, AX
MOV AX, DX
DIV BL
INT 3

```

Output:

AX=0002 BX=0006
CX=0000 DX=000C

Result Verification:

$(4! / 2!) / 3! = (24/2) / 6 = 12/6 = 2 = 2H = 02$

Discussion:

At first, 1 replaced in AX register and 4 replaced in CL register then do multiply by giving loop with CL address and move AX value in DX register. Again, entered value 1 in AX register and 2 replaced in CL register then do multiply by giving loop with CL address and move AX value in BX register. Then move the DX value in AX register and do division by BL. Now, move AX value in DX and again entered value 1 in AX register and 3 replaced in CL register then do multiply by giving loop with CL address. Now, we move the AX value in BX register then move the DX value in register AX and divide the value by BL. After pressing STP and REG, we get the result.

Problem 10: Byte with Byte Division

```
ORG 100h
.MODEL SMALL
.DATA
NUM_1 DB 0F2H
NUM_2 DB 4H
.CODE
MOV BH, NUM_2    ;Load numerator in BH
MOV AL, NUM_1    ;Load denominator in AL
DIV BH           ;Divide BH by AL
RET
```

Output:

AX=023C

The DIV instruction divides BH by AL. F2 divided by 04 gives quotient of 3C and give 02 as a remainder. AL stores the quotient and remainder is stored in AH register.

- ORG (abbr. for ORiGin) is an assembly directive (not an instruction). It defines where the machine code (translated assembly program) is to place in memory. As for ORG 100H this deals with 80x86 COM program format (COMMAND) which consist of only one segment of max. 64k bytes. 100H says that the machine code starts from address (offset) 100h in this segment, effective address is CS:100H.
- With .model small you get a program where CS points to a 64k bytes code segment and DS point to 64k bytes data segment. Thus, code and data both use 64k bytes maximum space.

```
.MODEL MEDIUM    ;the data must fit into 64K bytes
                  ;but the code can exceed 64K bytes of memory
.MODEL COMPACT    ;the data can exceed 64K bytes
                  ;but the code cannot exceed 64K bytes
.MODEL LARGE      ;both data and code can exceed 64K
                  ;but no single set of data should exceed 64K
.MODEL HUGE       ;both code and data can exceed 64K
                  ;data items (such as arrays) can exceed 64K
.MODEL TINY       ;used with COM files in which data and code
                  ;must fit into 64K bytes
```

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Problem 11: Word with Word Division

```
ORG 100h
.MODEL SMALL
.DATA
NUM_1 DW 0F213H
NUM_2 DW 41A8H
.CODE
```

MOV AX, NUM_1 ;Load numerator in AX DIV NUM_2 ;Divide AX by NUM_2 RET	
<u>Output:</u> The output window shows that the division of F213H by 41A8 gives the remainder of 2D1B into DX register and 03 as a quotient into AX.	AX=0003 DX=2D1B

Conclusion:

In this experiment, we have learnt conversion from °C to °F, conversion from °F to °C, conversion from °C to °K, Conversion from °K to °C, Average of 3 numbers, average of 5 numbers, area of rectangle, area of triangle, find how many tiles. After performing those operation, we use assembly language in 8086 microprocessors which results in getting the correct output.

Example: HomeWorks

1. Temperature conversion from °C to °F (37°C)
2. Temperature conversion from °F to °C (110°F)
3. Temperature conversion from °F to °K (130°F) ; $AX=0547H$; $K=(F-32)*5/9+273$;
4. Temperature conversion from °K to °F (300°K); $F=9(K-273)/5+32$
5. $3! + 4!$
6. $(4! + 3!) - 2!$
7. $(1! * 2!) * 6!$
8. Find out the average of the ten numbers.
9. Factorial Operation: $7! - 4! + 2!$
10. Floor size 80*80, Tiles size 4*4. How many tiles will be required to pave up the floor?